Prof. Bao-Shan Wang, Ph.D.

wbs@ouc.edu.cn
bao-shan.github.io
0009-0007-8744-0648
School of Mathematical Sciences, Ocean University of China (OUC)
No. 238, Songling Road, Qingdao 266100, China.
Google Scholar, ResearchGate, Web of Science



Academic Background

Sept., 2018 – June, 2022
- **Ph.D.** in Computational Mathematics, OUC.
- Prof. Wai Sun Don
- High-Order Physical-Constraint-Preserving WENO Schemes.

Sept., 2015 – June, 2018
- **M.Sc.** in Computational Mathematics, OUC.
- Prof. Zhen Gao and Prof. Wai Sun Don
- Radial Basis Function based Discontinuities Detection Algorithm and its Application in High Order Hybrid Scheme.

Aug., 2011 – June, 2015
- **B.Sc.** in Mathematics and Applied Mathematics, OUC.

Academic Experience

Sept., 2024 – present
- **Associate Professor**, School of Mathematical Sciences, OUC

Aug., 2022 – Sept., 2024
- **Postdoc**, School of Mathematical Sciences, OUC
- Prof. Zhen Gao
- High-Order Physical-Constraint-Preserving AWENO Scheme for Multi-Medium Flows.

Academic Visiting

2026
- Hong Kong Baptist University (HKBU), Jan. 4 – 15, Sept. 1 – 19
- Prof. Leevan Ling & Prof. Wai Sun Don @ Department of Mathematics

2025
- Pohang University of Science and Technology (POSTECH), Dec. 25, 2025 – Jan. 2, 2026
- Prof. Jae-Hun Jung & Dr. Jiaxi Gu @ Department of Mathematics

2025
- HKBU, Nov. 2 – Nov. 29
- Prof. Leevan Ling & Prof. Wai Sun Don @ Department of Mathematics

2024
- HKBU, Nov. 3 – Nov. 30
- Prof. Leevan Ling & Prof. Wai Sun Don @ Department of Mathematics

2024
- Southern University of Science and Technology (SUSTech), Match, 3 – May, 31
- Prof. Alexander Kurganov @ International Center for Mathematics

2024
- The Hong Kong University of Science and Technology, May, 5 – 11
- Prof. Zhichao Peng @ Department of Mathematics

2023
- HKBU, Oct. 16 – Nov. 6, Dec. 16, 2023 – Jan. 9, 2024
- Prof. Leevan Ling & Prof. Wai Sun Don @ Department of Mathematics

2023
- SUSTech, April, 3 – 9
- Prof. Alexander Kurganov @ Department of Mathematics

Academic Visiting (continued)

2019 SUSTeach, Mar. 27 – 31, Sept. 15 – 19, Dec. 17 – 20
Prof. Alexander Kurganov @ Department of Mathematics

Research Publications (Sum of Times Cited = 496, H-Index = 13)

($*$ = corresponding author; $\sharp$ = contributed equally; IF = impact factor)

1. Y.-R. Zhao, Z. Gao, **B.-S. Wang***, & W. S. Don. High-order physical-constraints-preserving common-weights A-WENO scheme for non-conservative five-equation model with stiffened gas EOS. *J. Comput. Phys.*, 2026, 115530, Accepted. (IF = 3.9)
2. Q. Fu, A. Kurganov*, & **B.-S. Wang**. Local characteristic decomposition-based central-upwind flux splitting. *J. Sci. Comput.*, 2026, Accepted. (IF = 2.6)
3. **B.-S. Wang***. On wave speeds in the HLLC Riemann solver for the compressible two-medium flows with general equations of state. *J. Comput. Phys.*, 2027, 569(2), 115501. (IF = 3.9)
4. K. Bensayah, **B.-S. Wang***, J.-H. Liu, & A. Hadjadj. WENO-ZKB: An improved WENO-Z+ scheme with new anti-dissipative term. *J. Comput. Phys.*, 2027, 569(1), 115434. (IF = 3.9)
5. Y. Qiao$^\sharp$, Y.-X. Liu$^\sharp$, **B.-S. Wang**$^{*,\sharp}$, & W.S. Don$^\sharp$. Affine-invariant trigonometric WENO finite difference schemes for hyperbolic conservation laws. *Commun. Comput. Phys.*, 2027, 41(3), 944–983. (IF = 3.17)
6. C. Guo$^\sharp$, **B.-S. Wang**$^\sharp$, W.S. Don$^\sharp$, & Y. Qiao$^{*,\sharp}$. High-order physical-constraints-preserving velocity-consistent schemes for compressible multicomponent five-equation model with the Mie-Grüneisen equation of state. *J. Comput. Phys.*, 2026, 563, 115049. (IF = 3.9)
7. Y.-R. Zhao, Z. Gao, & **B.-S. Wang***. Fifth-order equilibrium-preserving path-conservative characteristic-wise AWENO scheme for one-fluid two-temperature Euler model. *J. Comput. Phys.*, 2026, 544, 114409. (IF = 3.9)
8. C. Guo$^\sharp$, Y. Qiao$^\sharp$, B.-S. Wang$^\sharp$, & W.S. Don$^{*,\sharp}$. Bound- and positivity-preserving first-order velocity-consistent HLL scheme for two-medium γ-based model of stiffened gas. in: D. E. Gildfind, R. Gollan and H. H. Kleine (eds), *Proceedings of the 35th International Symposium on Shock Waves: ISSW35, July 7-11 2025, Brisbane, Australia*.
9. X.-S. Xiang, P. Li, & **B.-S. Wang***. Finite-volume WENO scheme on non-uniform mesh for 3D detonation simulation. *Appl. Math. Let.*, 2026, 182, 110046. (IF = 2.7)
10. **B.-S. Wang**, A.S.L. Chu, L. Ling, & W.S. Don*. Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds: I. Geometry-Compatible Fluxes and Numerical Validation on Sphere and Torus. *Commun. Nonlinear Sci. Numer. Simul.*, 2026, 152, 109415. (IF = 3.8)
11. X.-S. Xiang, P. Li, Z. Gao, & **B.-S. Wang***. High-order well-balanced positivity-preserving affine-invariant finite volume CWENOZ scheme with LDCU flux for the Euler equations under gravitational fields. *Appl. Numer. Math.*, 2026, 224, 181–206. (IF = 2.4)
12. K.-B. Tian, **B.-S. Wang**, X. Wen, & Z. Gao*. High order well-balanced and positivity-preserving AWENO scheme for rotating shallow water equations with Coriolis force. *East Asian J. Appl. Math.*, 2026, 16(4), 714–741.

13. W. S. Don♯, J.-L. Li♯, L. Ling♯, **B.-S. Wang***,♯, & Y. Wang♯. Hybrid high-order shock-capturing scheme for one-dimensional hyperbolic conservation laws on manifolds (surface PDEs) in the time-continuous embedding framework. in: S. Chun, J.-H. Jung, E.J. Park, J. Shen (eds), *Spectral and High-Order Methods for Partial Differential Equations ICOSAHOM 2023. Lecture Notes in Computational Science and Engineering*, vol. 142, Springer, Cham, 2025, pp. 239–255.

14. **B.-S. Wang** & W.S. Don*. Fifth-order bound-, positivity-, and equilibrium-preserving affine-invariant AWENO scheme for two-medium γ-based model of stiffened gas. in: R.S. Myong and H.D. Kim (eds.), *Proceedings of the 34th International Symposium on Shock Waves, Volume 1: Fundamentals*, 2025, pp. 59-71.

15. C.-F. Wang♯, W.S. Don♯, J.-L. Li♯, & **B.-S. Wang***,♯. Improved sixth-order WENO finite difference schemes for hyperbolic conservation laws. *Adv. Appl. Math. Mech.*, 2025, 17(6), 1591-1624. (IF = 1.1)

16. Q. Fu, Y. Gu*, A. Kurganov, & **B.-S. Wang**. Bound- and positivity-preserving path-conservative central-upwind AWENO scheme for the five-equation model of compressible two-component flows. *J. Sci. Comput.*, 2025, 104(3), 94. (IF = 3.3)

17. C. Guo, Y. Qiao, W.S. Don, & **B.-S. Wang***. A family of bound-preserving velocity-consistent schemes for two-medium γ-based model with stiffened gas. *J. Comput. Phys.*, 2025, 538, 114149. (IF = 3.8)

18. J.-L. Li♯, W.S. Don♯, C.-F. Wang♯, & **B.-S. Wang***,♯. Spatial-temporal adaptive-order positivity-preserving WENO finite difference scheme with relaxed CFL condition for Euler equations with extreme conditions. *Adv. Appl. Math. Mech.*, 2025, 17(3), 804–839. (IF = 1.5)

19. Z. Gao, Z.-Y. Tang, **B.-S. Wang***, & Y.-R. Zhao. Novel high-order alternative finite difference central WENO schemes for hyperbolic conservation laws. *Numer. Meth. Part. Diff. Eq.*, 2025, 41(2), e23170. (IF = 2.1)

20. X.-S. Xiang♯, **B.-S. Wang**♯, P. Li*,♯, & Z. Gao♯. Affine-Invariant WENO Operator on Nonuniform Cartesian Mesh with Application to Finite Volume and Discontinuous Galerkin Methods. *J. Sci. Comput.*, 2025, 102(2), 29. (IF = 2.8)

21. Z. Gao♯, S. Guo♯, **B.-S. Wang**♯, & Y. Gu*,♯. High order bound- and positivity-preserving finite difference affine-invariant AWENO scheme for the five-equation model of two-medium flows. *Commun. Comput. Phys.*, 2024, 36(3), 781–820. (IF = 2.6)

22. B. Ren, **B.-S. Wang**, X. Zhang, & Z. Gao*. Fourth order positivity and bound preserving well-balanced compact finite difference scheme for Ripa and pollutant transport systems. *Comput. Math. Appl.*, 2024, 176, 545–563. (IF = 2.9)

23. **B.-S. Wang** & N.K. Garg*. Third-order numerical scheme for Euler equations of gas dynamics using Jordan canonical based splitting flux. *Comput. Fluids*, 2024, 281, 106370. (IF = 2.5)

24. Z. Gao, Y.-R. Zhao, & **B.-S. Wang***. Sixth-order perturbed WENO interpolation-based AWENO and WCNS-E schemes for hyperbolic conservation laws. *Appl. Math. Let.*, 2024, 158, 109230. (IF = 2.9)

25. Q. Fu♯, Z. Gao♯, Y. Gu♯, P. Li*,♯, **B.-S. Wang**♯. Improved Well-Balanced AWENO Schemes with Hydrostatic Reconstruction for the Euler Equations Under Gravitational Fields. *Math. Comput. Simul.*, 2024, 221, 260–280. (IF = 4.6)

26 P. Li♯, T. Li♯, W.S. Don♯, **B.-S. Wang***,♯. Scale-invariant multi-resolution alternative WENO scheme for the Euler equations. *J. Sci. Comput.*, 2023, 94(1), 15. (IF = 2.5)

27 **B.-S. Wang**, W.S. Don, P. Li*. Fifth-order well-balanced positivity-preserving finite difference AWENO scheme with hydrostatic reconstruction for hyperbolic chemotaxis models. *Appl. Numer. Math.*, 2023, 186, 41–56. (IF = 2.994)

28 Y. Wang, W.S. Don, **B.-S. Wang***. Fifth order AWENO finite difference scheme with adaptive numerical diffusion for Euler equations. *Comput. Fluids*, 2023, 251, 105743. (IF = 3.077)

29 **B.-S. Wang**, W.S. Don, A. Kurganov*, Y. Liu. Fifth-order A-WENO schemes based on the adaptive diffusion central-upwind Rankine-Hugoniot fluxes. *Commun. Appl. Math. Comput.*, 2023, 5, 295–314. (IF = 1.6)

30 K.-B. Tian, W.S. Don, **B.-S. Wang***. High order WENO finite difference scheme with adaptive dual order ideal weights for hyperbolic conservation laws. *Appl. Numer. Math.*, 2023, 187, 50–70. (IF = 2.994. The 16th most downloaded article from Appl. Numer. Math. in the last 90 days, Oct. 25th, 2023.)

31 Y. Wang♯, **B.-S. Wang**♯, L. Ling♯, W.S. Don*,♯. A time-continuous embedding method for scalar hyperbolic conservation laws on manifolds. *J. Sci. Comput.*, 2022, 93(3), 84. (IF = 2.5)

32 **B.-S. Wang**, W.S. Don*. Affine-invariant WENO weights and operator. *Appl. Numer. Math.*, 2022, 181, 630–646. (IF = 2.994)

33 **B.-S. Wang**, P. Li, Z. Gao*. High order well-balanced and positivity-preserving finite difference AWENO scheme with hydrostatic reconstruction for shallow water equations. *Appl. Numer. Math.*, 2022, 181, 483–502. (IF = 2.994)

34 W.S. Don♯, R. Li♯, **B.-S. Wang**♯, .*,♯Y. Wang A novel and robust scale-invariant WENO scheme for hyperbolic conservation laws. *J. Comput. Phys.*, 2022, 448, 110724. (IF = 4.645)

35 P. Li♯, **B.-S. Wang**♯, W.S. Don*,♯. Sensitivity parameter-independent characteristic-wise well-balanced finite volume WENO scheme for the Euler equations under gravitational fields. *J. Sci. Comput.*, 2021, 88(2), 47. (IF = 2.843)

36 Z. Gao*, Q. Liu, J.S. Hesthaven, **B.-S. Wang**, W.S. Don, X. Wen. Non-intrusive reduced order modeling of convection dominated flows using artificial neural networks with application to Rayleigh-Taylor instability. *Commun. Comput. Phys.*, 2021, 30(1), 97–123. (IF = 3.246)

37 **B.-S. Wang**, W.S. Don, N.K. Garg*, A. Kurganov. Fifth-order A-WENO finite difference schemes based on a new adaptive diffusion central numerical flux. *SIAM J. Sci. Comput.*, 2020, 42(6), A3932–A3956. (IF = 2.373)

38 W.S. Don, D.-M. Li, Z. Gao, **B.-S. Wang***. A characteristic-wise alternative WENO-Z finite difference scheme for solving the compressible multicomponent non-reactive flows in the overestimated quasi-conservative form. *J. Sci. Comput.*, 2020, 82(2), 27. (IF = 2.592)

39 Z. Gao, L.-L. Fang, **B.-S. Wang**, Y. Wang, W.S. Don*. Seventh and ninth orders alternative WENO finite difference schemes for hyperbolic conservation laws. *Comput. Fluids*, 2020, 202, 104519. (IF = 3.013)

40 Y. Yang, W.S. Don, Z. Gao, **B.-S. Wang***. Hybrid compact-WENO scheme with RBF-FD based discontinuity detection method for hyperbolic conservation laws. *J. Numer. Meth. Comput. Appl.*, 2020, 41(3), 232–245.

41. J. Xu, Z. Gao, W.S. Don, **B.-S. Wang***. Order-preserving sixth-order WENO finite difference scheme in the presence of high order critical points. *J. Numer. Meth. Comput. Appl.*, 2020, 41(1), 68–82.

42. L. Wang*, Z. Chen, et al. Review of hydrodynamic instabilities in inertial confinement fusion implosions. *High Power Laser and Particle Beams*, 2020, 33(1), 012001.

43. P. Li, X. Zhao, Z. Gao*, **B.-S. Wang**. High order hybrid weighted compact nonlinear schemes for hyperbolic conservation laws. *Adv. Appl. Math. Mech.*, 2020, 12(4), 972–991. (IF = 1.727)

44. Y. Wang♯, **B.-S. Wang**♯, W.S. Don*,♯. Generalized sensitivity parameter free fifth order WENO finite difference scheme with Z-type weights. *J. Sci. Comput.*, 2019, 81(3), 1329–1358. (IF = 2.228)

45. **B.-S. Wang**, W.S. Don*, Z. Gao, Y. Wang, X. Wen. Hybrid Compact-WENO finite difference scheme with radial basis function based shock detection method for hyperbolic conservation laws. *SIAM J. Sci. Comput.*, 2018, 40(6), A3699–A3714. (IF = 2.310)

46. **B.-S. Wang**, P. Li, Z. Gao, W.S. Don*. An improved fifth order alternative WENO-Z finite difference scheme for hyperbolic conservation laws. *J. Comput. Phys.*, 2018, 374, 469–477. (IF = 2.845)

47. **B.-S. Wang**, W.S. Don, Z. Gao*, Y.-H. Chen. Vortex identification based multi-resolution analysis for hybrid compact-WENO scheme. *Periodical of Ocean University of China*, 2018, 48(S2), 198–202.

48. W.S. Don♯, **B.-S. Wang**♯, Z. Gao*,♯. Fast iterative adaptive multi-quadric radial basis function method for edges detection of piecewise functions—I: uniform mesh. *J. Sci. Comput.*, 2018, 75(2), 1016–1039. (IF = 2.370)

Articles under Review/Preparation

1. S. Chu, A. Kurganov*, M. Na, **B.-S. Wang**, & R. Xin. Local characteristic decomposition of equilibrium variables for hyperbolic systems of balance laws. *Appl. Numer. Math.*, under revision.

2. Y.-R. Zhao, Z. Gao, & **B.-S. Wang***. Physical-constraints-preserving path-conservative Lax-Friedrichs schemes for non-conservative two-medium six-equation model. *J. Sci. Comput.*, under review.

3. A. S. L. Chu♯, W. S. Don♯, L. Ling♯, & **B.-S. Wang***,♯. Foliation structures and global flow dynamics of scalar hyperbolic conservation laws on manifolds: II. An adaptively penalized closest point embedding with a WENO finite difference scheme (AP-cp-WENO). submitted to *SIAM J. Sci. Comput.*

4. J.-H. Liu, **B.-S. Wang**, & Y.-R. Zhao*. Sixth-order AWENO schemes with adaptive linear weights for hyperbolic conservation laws. submitted to *Comput. Fluids*.

5. C. Guo, P. Huang*, Y. Qiao, **B.-S. Wang**. Accurate simulation of the granular avalanche flows by high-order positivity-preserving affine-invariant AWENO finite difference schemes. submitted to *Comput. Fluids*.

6. Y. Qiu, Z. Gao, A. Kurganov, **B.-S. Wang***, & X. Wen. Fifth-order well-balanced path-conservative A-WENO scheme for the Ripa model. submitted to *J. Comput. Math.*

7. M. Zhang♯, **B.-S. Wang**♯, W. S. Don♯, & Y. Qiao*,♯. Spatially adaptive-order bound-preserving velocity-consistent AWENO schemes for compressible two-medium flows. submitted to *J. Sci. Comput.*
8. H.-L. Nie, Y. Qiao, **B.-S. Wang***, & W. S. Don. RBF-AWENO scheme with affine-invariant weights: Achieving sixth-order accuracy and multiscale robustness. *Commun. Comput. Phys.*, under revision.
9. N. K. Garg*, J. Pal, & **B.-S. Wang**. A Low-Dissipation LLF-Type Adaptive Central Scheme for the Compressible Euler Equations. submitted to *Math. Comput. Simulat.*
10. J. Gu, **B.-S. Wang***, W. S. Don, & J.-H. Jung. A family of even-order central-upwind WENO schemes with averaged downwind and novel global smoothness indicators. submitted to *Numer. Meth. Part. Diff. Eq.*
11. J. Li, P. Li* & **B.-S. Wang**. Physical-constraints-preserving conservative-variable AWENO scheme for two-medium flows. submitted to *Adv. Appl. Math. Mech.*

Books in Preparation

1. W.S. Don, **B.-S. Wang**. Lecture on the High-Order WENO Scheme for Hyperbolic Conservation Laws. DOI: 10.13140/RG.2.2.24752.37123/2.
2. Wai Sun Don & **Bao-Shan Wang**. Comprehensive Lecture Notes on Hyperbolic Conservation Laws on Manifolds. DOI:
3. **B.-S. Wang**. My Mathematical Experiments on MATLAB.

Research Grants

Jan., 2025 – Dec., 2027	PI, startup funding at OUC, 300K.
Jan., 2024 – Dec., 2026	PI, NSFC, 12301530, 300K.
July, 2023 – July, 2024	PI, China Postdoctoral Science Foundation, 2023M733348, 80K.

Teaching Experiences

Feb, 2025 – July, 2025	Complex Function.
Feb, 2025 – June, 2025	Spectral Methods.
Sept., 2017 – Aug, 2023	Research Skills Training (MATLAB, Fortran, LaTeX, Linux).
Sept., 2021 – Jan., 2022	Mini-Courses: Scientific Computing and Programming on MATLAB, Nonlinear Wave Equations.
Mar., 2021 – July, 2021	Teaching: Spectral Methods for Time–Dependent Problems.
Sept., 2015 – July, 2016	Teaching Assistant: Mathematical Experiments on MATLAB.

Conferences/Seminars/Talks

Oct., 2026	Spatially Adaptive-Order Bound-Preserving Velocity-Consistent AWENO Schemes for Compressible Two-Medium Flows, High-Order and Adaptive Schemes for Nonlinear Hyperbolic PDEs, SUSTech, Shenzhen, 17 Oct.

Conferences/Seminars/Talks (continued)

- Aug., 2026 — Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds, EASIAM 2026, Jeju, South Korea, 27 Aug.
- Aug., 2026 — High-order physical-constraints-preserving velocity-consistent schemes for compressible multicomponent five-equation model with the Mie-Grüneisen equation of state, Kunming, Yunan, 18 Aug.
- Jan., 2026 — High-Order Well-Balanced AWENO Scheme for Extended Shallow Water Models, Laoshan Laboratory, Qingdao, 30 Jan.
- Jan., 2026 — Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds, Workshop on High-order Numerical Methods for Partial Differential Equations with Applications, Nanning, 15-18 Jan.
- Jan., 2026 — Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds, Conference on Modern Kernel Methods and Applications, On the Occasion of Robert Schaback's 80th Birthday, HKBU, 5-9 Jan.
- Dec., 2025 — A family of bound-preserving velocity-consistent schemes for two-medium γ-based model with stiffened gas, POSTECH, Pohang, Korea
- Oct., 2025 — A family of bound-preserving velocity-consistent schemes for two-medium γ-based model with stiffened gas, CSIAM 2025, Changsha, Hunan
- Aug., 2025 — A family of bound-preserving velocity-consistent schemes for two-medium γ-based model with stiffened gas, 21st NSNMF, Yining, Xinjiang
- July, 2025 — Bound- and Positivity-preserving Path-Conservative Central-Upwind AWENO Scheme for the Five-equation Model of Compressible Two-component Flows, Xinjiang University, Urumqi
- Dec., 2024 — Bound- and Positivity-preserving Path-Conservative Central-Upwind AWENO Scheme for the Five-equation Model of Compressible Two-component Flows, Henan University of Technology, Zhengzhou
- Nov., 2024 — Time-Continuous Embedding Method for Hyperbolic Conservation Laws on Manifolds, HKBU, Hong Kong
- June, 2024 — Fifth-order bound-, positivity-, and equilibrium-preserving affine-invariant AWENO scheme for two-medium γ-based model of stiffened gas, Xinjiang University, Urumqi, China
- April, 2024 — Time-Continuous Embedding Method for Scalar Hyperbolic Conservation Laws on Manifolds, Shijiazhuang Tiedao University, Shijiazhuang, China
- Mar., 2024 — Affine-invariant WENO operator and applications in solving hyperbolic conservation laws, Workshop on Development of High-Order Methods for Hyperbolic PDEs, ICM, SUSTech, Shenzhen, China
- Nov., 2023 — Time-Continuous Embedding Method for Scalar Hyperbolic Conservation Laws on Manifolds, Harbin Institute of Technology, Shenzhen, China
- Oct., 2023 — Affine-invariant WENO weights and applications in solving hyperbolic conservation laws, HKBU, Hong Kong.
- Aug., 2023 — Affine-invariant WENO weights and operator, ICOSAHOM 2023, Korea.
- Apr., 2023 — Affine-invariant WENO weights and operator, SUSTech, Shenzhen, China.
- 20th National Symposium on Numerical Methods in Fluids, Nanjing, China.

Conferences/Seminars/Talks (continued)

July, 2022 — High Order Affine-Invariant WENO Scheme, Xinjiang University (Online).
July, 2021 — Workshop on Shandong Computational Mathematics 2021, Yantai University.
Nov., 2020 — China SIAM Student Forum (Online).
Apr., 2020 — Workshop on Applied and Computational Mathematics, Shijiazhuang, China.
Nov., 2019 — Int'l Joint Conf. on AI & Data Science: Math. & Appl., Suwon, Korea.
June, 2019 — 11th Int'l Conf. on Computational Physics, Hangzhou, China.
Dec., 2018 — Numerical Simulation on Ablative Rayleigh-Taylor Instability, IAPCM, Beijing.
Aug., 2017 — 18th National Symposium on Numerical Methods in Fluids, Huaihua, China.

Professional Services

Journal Reviewer (in part)

- J. Comput. Phys., Appl. Numer. Math., Commun. Comput. Phys., Comput. Fluids
- Commun. Nonlinear Sci. Numer. Simul., Comput. Math. Appl., Math. Comput. Simul.
- East Asian J. Appl. Math., Adv. Appl. Math. Mech.

Mini-symposium Organizer

- EASIAM 2026, Aug. 23–27, 2026, Jaju, Korea.
- ICOSAHOM 2023, Aug. 14–18, 2023, Yonsei University, Seoul, Korea.

Memberships

- China SIAM member.

Honors/Awards

2023 — Outstanding PhD Degree Dissertation of Shandong Province.
2021 — China National Scholarship for Doctoral Students.
2021 — Eighth Annual Graduate Character at OUC (10 persons/groups in total).
2019 — Outstanding Master Degree Dissertation of Shandong Province.
2018 — Outstanding Master Degree Graduates of Shandong Province.
2017 — China National Scholarship for Graduate Students.
2016 — National Post-Graduate Mathematical Contest in Modeling: First Prize.
2015 — National Post-Graduate Mathematical Contest in Modeling: Second Prize.
2014 — China Undergraduate Mathematical Contest in Modeling: Second Prize.

Skills

- Languages: Chinese, English.
- Operating systems: Linux, Windows.
- Programming: Fortran, MATLAB.
- Softwares: LaTeX, Office softwares.
- High performance parallel scientific computing: CUDA, MPI, OpenACC, OpenMP.